# 25. THE RENAL SYSTEM : GLOBAL EMBRYONIC KINETICS

DURATION : 06:56 STUDY SHEET

### COURSE SUMMARY

This video explores in depth the embryonic kinetics of the renal system, starting with its mesodermal origin, then illustrating how flexion and endorotation forces influence kidney development. Key concepts include the cranio-caudal development of the different stages of the kidney – Pronephros, Mesonephros, and Metanephros – and the dynamics of compression and rotation movements within the mesoderm. Therapists will learn to recognise how these embryonic mechanisms establish the foundations of organ systems, thus offering a better understanding of human anatomy and physiology.

## THE RENAL SYSTEM: GLOBAL EMBRYONIC KINETICS

### 1. THE MESODERMAL ORIGIN OF THE RENAL SYSTEM

The lesser pelvis and the **renal system** are intrinsically linked in their development.

The establishment of the renal system begins at the **mesodermal** level, from the primitive streak. We find there:

- **The somatic plate**: This will give rise to the somites, which then differentiate into **sclerotome**, **myotome**, and **dermatome**.
- **The lateral plate**: This will later form the **somatopleure** and the **splanchnopleure**.
- **The intermediate plate**: It is in this tissue that the renal system develops. It will give rise to the **excretory** and **urogenital tissue**. From this stage, all the potential is present.

The kidney is a tissue formed by compression resulting from flexion, which is a concentration of **intermediate mesoderm**.

### 2. FLEXION AND ENDOROTATION: THE SCULPTING FORCES

At the embryonic stage, we observe:

- The **neural tube**.
- The **notochord**.

- The **somites** (dermatome, sclerotome).
- The **aortas**, which are very important.

The intermediate plate, initially open, closes on its longitudinal axis.

The growth of the neural tube and the approximation of the aortas lead to:

- A movement of **longitudinal compression**.
- A movement of **endorotation** in the longitudinal axis (a torsion).

The tissue, in its embryonic memory, receives compression and a rotational movement. It is a tissue that revitalises through compression.

This first biodynamic information of developmental kinetics is a concentration of mesoderm generated by flexion and this movement of approximation.

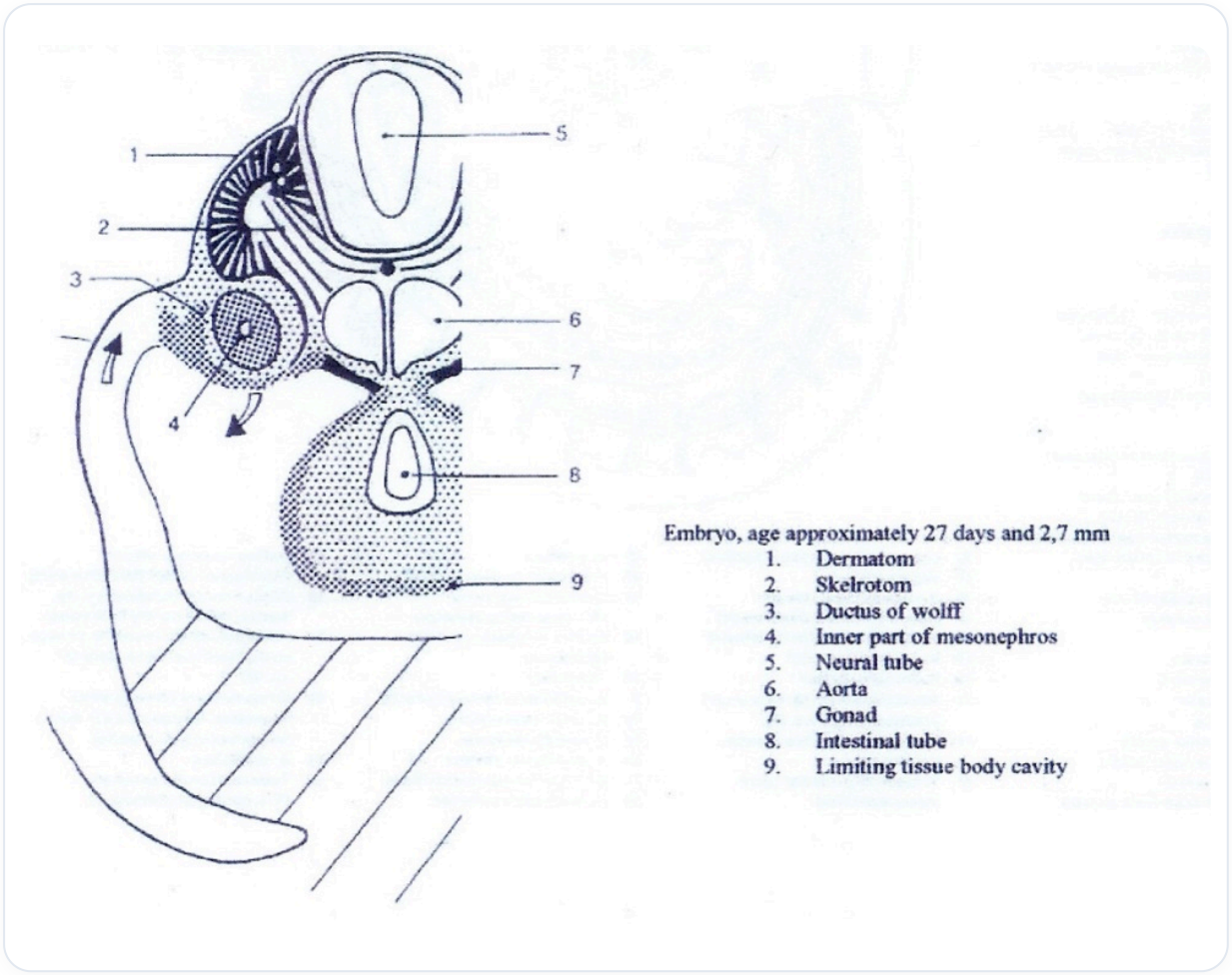



*FIG. — Concept of kidney formation by compression and flexion*

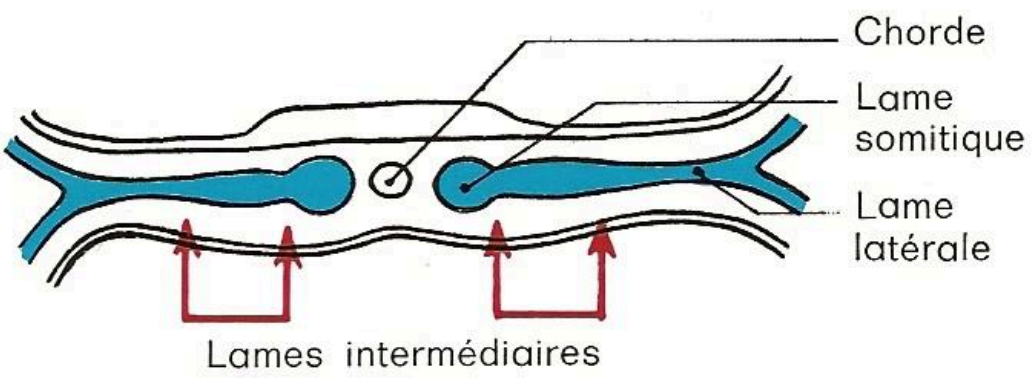



Fig. 1. — *Schéma de l'embryon, plan tridermique.*

*FIG. — Tridermic plan embryo*

The posterior intermediate plate of the posterior parietal peritoneum also undergoes this movement.

In this ridge, a canal appears: the **mesonephric duct**, a collecting duct in formation.

## 3. EMBRYONIC ELONGATION AND MIDLINE ROTATION

At the beginning of the fifth week, the embryo elongates. A larger fold is observed at the level of the **coelom**.

From a frontal view, these two large folds reveal a **rotation** organised by the approximation of the mesoaortas on the midline.

There is therefore **compression and rotation**, this is the first major information in the mesoderm. This indicates a repositioning of the system that we will detail.

## 4. THE CRANIO-CAUDAL DEVELOPMENT OF THE KIDNEY

The kidney, in its developmental phase, builds from **cranial** to **caudal**. Three main stages are distinguished:

1. **The Pronephros**:

* This is the first part of the kidney.

* It develops at an almost cervical level.

* It quickly emits a small collecting duct and then disappears.

2. **The Mesonephros**:

* This is the second part that develops.

* It becomes a functional kidney, forming a connection with the aorta and the first **glomeruli**.

* It also disappears after a certain time.

3. **The Metanephros**:

* This is the third part that develops.

* It is induced by a **ureteric bud**.

* It will form the definitive kidney.

Each part allows the emergence of elements that will be incorporated into the definitive kidney.

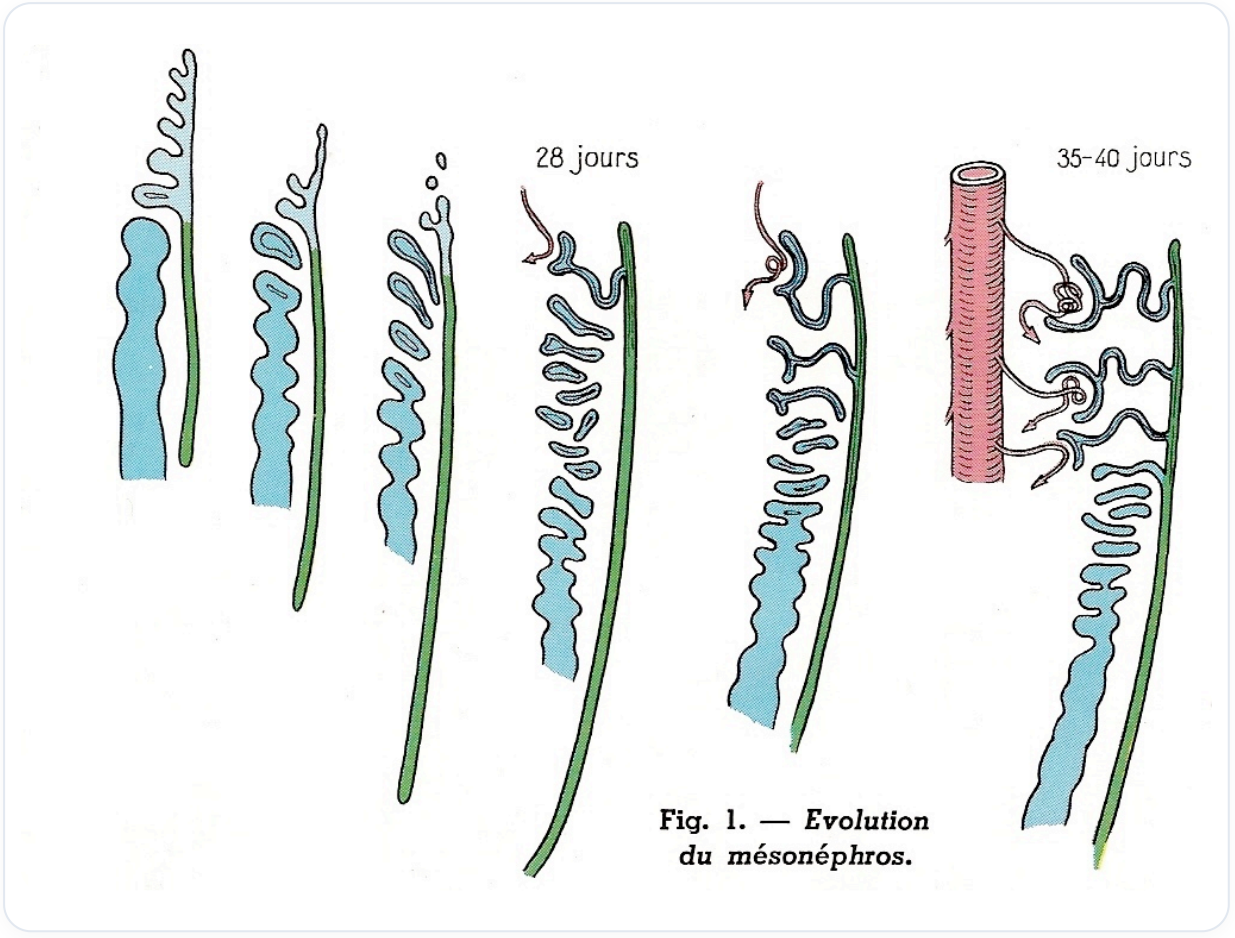



*FIG. — Développement crânio-caudal du rein (Pronéphron, Mésonéphron, Métanéphron) qui est le sujet du paragraphe*

This is a **cranio-caudal** developmental sequence. The first impulse movement of the intermediate plate is compression, due to the differential growth between the neural tube and the aorta, which compresses the system.

## 5. REPOSITIONING AND DIFFERENTIAL GROWTH

Initially, the **adrenal glands** are enormous and follow this phase.

However, the kidneys do not ascend. It is the posterior part that descends faster.

The **sacrum** and the kidney become reference points. The kidney is the **fulcrum** for the development of the posterior part.

There is a cranio-caudal developmental sequence: Pronephros, Mesonephros, Metanephros. They do not appear at the same time. Compression occurs from top to bottom.

The primitive kidney gives rise to a small primitive collecting duct, which will be incorporated into the second kidney, the mesonephros, with a mesonephric duct.

Then, the third part will be the definitive kidney, called the metanephric kidney.

Initially, the mesodermal plate is undifferentiated. But to have the definitive kidney, this first small, cervical system is needed, which almost follows the path of a **meridian**. The **bladder** system will also develop from a small canal.

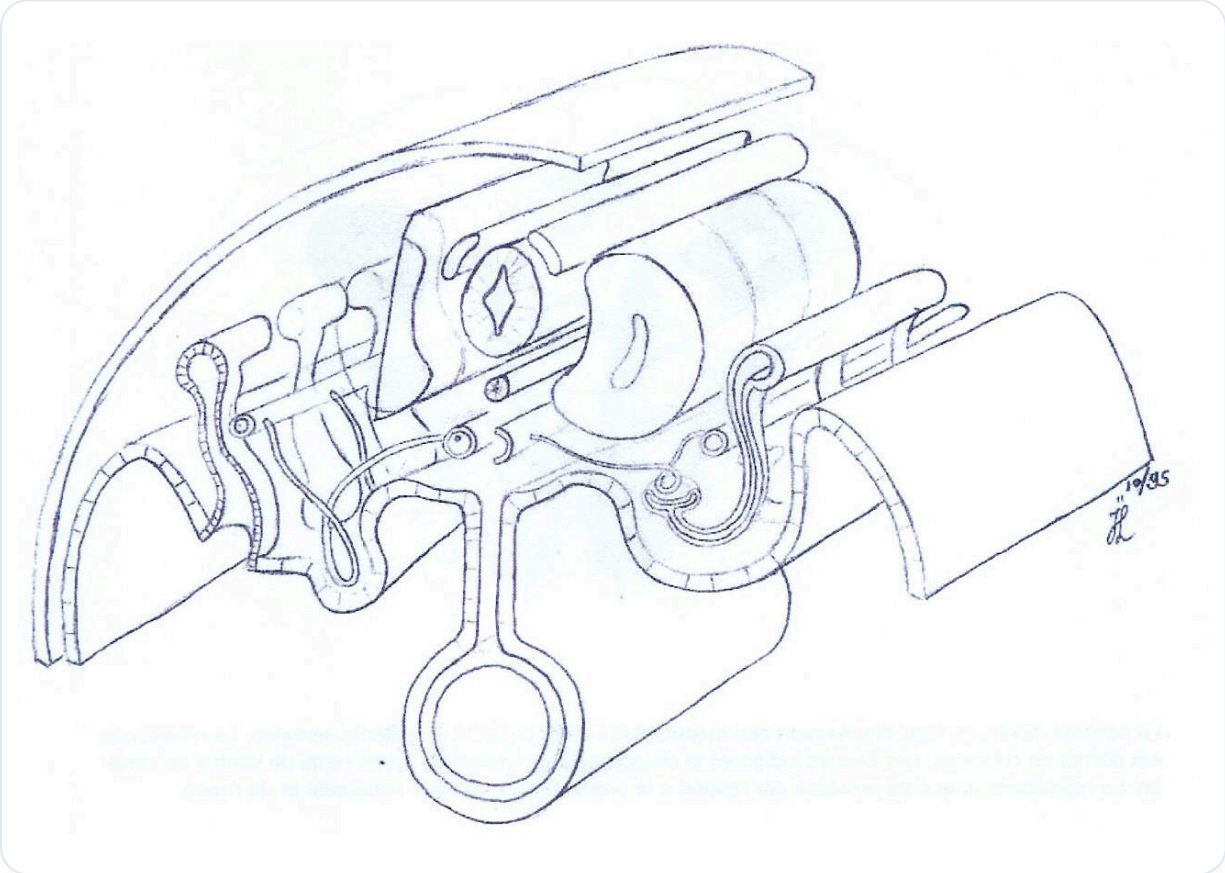


*FIG. — Vertebrate embryonic anatomy*